
CareClarity: A Safety-Bounded Multilingual LLM System for Improving Patient Understanding of UK NHS Administrative Letters

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Abstract: Patients in the UK regularly receive healthcare administrative letters about appointments, referrals, prescriptions, waiting lists and clinic changes. These documents can be difficult to understand, particularly for people with limited health literacy, accessibility needs or a preferred language other than English. General-purpose large language models may explain such letters, but can also create risk by drifting into diagnosis, treatment or medication advice. This paper presents CareClarity, a safety-bounded multilingual LLM system for explaining NHS-style administrative correspondence. CareClarity extracts key administrative details, rewrites complex wording in plain English, flags missing or ambiguous instructions, generates appointment-readiness checklists, supports family/carer summaries and translates letters into multiple patient languages. The system combines server-side LLM calls, structured JSON schemas, explicit admin-only prompting, safety validation, unsafe-output filters and fallback responses. We describe the system architecture and report a 100-document offline synthetic stress test of the deterministic fallback analyser. The benchmark produced 100/100 schema-valid outputs and 0/100 detected unsafe outputs, with 100.0% date/time accuracy, 79.0% location accuracy, 87.0% contact extraction and 100.0% valid action-checklist generation. Missing-detail detection was weaker in the stress test ($F1 = 0.357$), revealing false positives around referral, form and cancellation wording and motivating further LLM and multilingual evaluation.

Keywords: large language models, AI for healthcare, patient communication, trustworthy AI, natural language processing, multilingual health communication, human-centred AI

1. Introduction

Patients often interact with healthcare systems through written administrative communication: appointment letters, referral updates, prescription collection messages, waiting-list validation requests, cancellation notices and pre-appointment instructions. Although such letters are not clinical decision tools, misunderstanding them can still have practical consequences. Patients may miss appointments, attend the wrong site, fail to bring required documents, overlook pre-appointment tasks or remain unsure who to contact for clarification.

NHS England’s Accessible Information Standard requires health and adult social care services to identify, record, flag, share and meet relevant information and communication support needs [1]. NHS England’s community language translation and interpreting framework further highlights the need for consistent, high-quality language support for people with limited English proficiency [2]. The Healthcare Safety Investigation Branch has also reported safety risks around written communications in community languages, including

cases where appointment letters and pre-appointment instructions were not available in languages other than English [3]. These sources indicate a practical gap: many patients need assistance not with medical diagnosis, but with understanding what healthcare administration is asking them to do.

Large language models (LLMs) provide a promising interface for explaining complex text, but open-ended medical assistance is high risk. Prior work has shown that LLMs can encode clinical knowledge, while also emphasising that clinical applications require careful evaluation for factuality, harm and bias [6]. This motivates a narrower research question: can an LLM system be useful in healthcare by deliberately avoiding clinical advice and focusing only on administrative understanding?

This paper presents CareClarity, a prototype patient-facing system for explaining UK NHS-style administrative letters. The contribution is not a general medical chatbot. Instead, CareClarity is designed around a strict boundary: it explains administrative meaning, extracts actionable details and prompts users to contact appropriate NHS services, while refusing or falling back when a response crosses into diagnosis, treatment, prescribing or medication-change advice.

The main contributions are:

- a safety-bounded LLM design for patient-facing healthcare administration;
- a multilingual workflow for translating administrative letters while preserving critical details such as dates, times, locations and contact numbers;
- a structured output schema for appointment, referral, prescription-admin and waiting-list communications;
- a preliminary offline benchmark measuring extraction accuracy, missing-detail detection, safety compliance and schema validity on synthetic NHS-style letters, with an evaluation protocol for live LLM and multilingual testing.

2. Related Work and Literature Review

2.1. Patient communication and NHS administrative access

Clear patient communication is an established safety and accessibility issue. NHS England’s Accessible Information Standard requires services to make health and care information accessible to people with relevant communication needs [1]. The NHS community language translation and interpreting framework similarly emphasises consistent support for people with limited English proficiency [2]. HSSIB’s investigation into booking-system failures further shows that written appointment communications in community languages can create patient-safety risks when letters or instructions are not understandable to patients [3].

Missed appointments are also an operational concern for outpatient services, and NHS England guidance frames did-not-attend reduction as a way to improve patient experience and release elective-care capacity [4]. CareClarity does not claim to reduce missed appointments directly; it targets a plausible upstream factor: whether patients can identify the correct date, time, location, contact route and action required. This connects to wider plain-language communication literature, which argues that medical and healthcare information should be written in a form that patients can understand and act upon [5].

2.2. Large language models in healthcare

Recent healthcare LLM research shows both potential and risk. Med-PaLM demonstrated that large language models can encode clinical knowledge, but also showed the need for human evaluation across factuality, precision, potential harm and bias [6]. A 2025 systematic review of LLMs for patient care maps growing applications while stressing implementation challenges around safety, reliability and evaluation [7]. More recent evaluation work warns that benchmark choices can create an “evaluation illusion” in medicine, where reported performance may not translate to real clinical utility [8].

2.3. Trustworthy and safety-bounded AI

WHO guidance on AI for health and large multi-modal models emphasises governance, accountability, transparency and risk management [9, 10]. Broader LLM safety literature also warns that fluent language generation can mask unsupported, biased or harmful content [11]. CareClarity is positioned within this literature as a deliberately narrow administrative assistant rather than a diagnostic or treatment system. Its contribution is to explore whether task narrowing, structured schemas, validation and refusal/fallback behaviour can make an LLM-based tool useful for patient-facing NHS administration without claiming clinical authority.

Table 1. CareClarity task boundary.

Allowed admin-support behaviours	Disallowed medical behaviours
Explain appointment, referral, waiting-list and prescription-admin wording in plain English.	Diagnose a condition or infer what disease the user has.
Extract dates, times, locations, contact details, clinic names and actions.	Recommend treatment, tests, procedures or clinical decisions.
Flag missing or unclear administrative details.	Tell the user to start, stop, increase, decrease or change medication.
Generate safe questions to ask the GP practice, hospital team, pharmacist or admin office.	Tell the user to ignore, delay or overrule clinician instructions.
Translate administrative content into supported languages with safety notes.	Claim official NHS approval or endorsement.

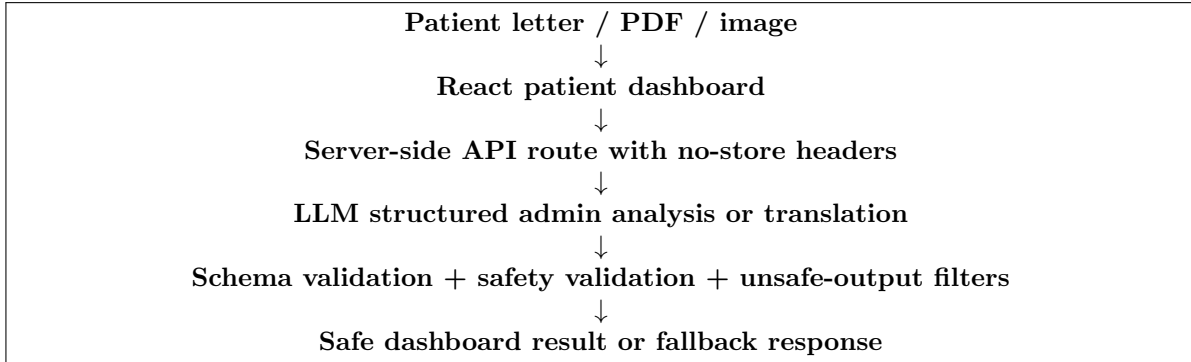


Figure 1. High-level CareClarity architecture. The API key is kept server-side and model outputs are validated before display.

3. System Scope and Design Requirements

CareClarity is designed for UK NHS-style administrative paperwork, including appointment letters, referral updates, prescription administration messages, waiting-list notices, clinic changes and jargon-heavy summaries. It is explicitly out of scope for diagnosis, treatment selection, medication changes, test interpretation or emergency triage.

The system requirements are: administrative usefulness, explicit safety boundaries, schema-valid structured output, multilingual access with critical-detail preservation, privacy-aware prototype design and conservative fallback behaviour when model output is invalid or unsafe.

4. CareClarity System Architecture

CareClarity is implemented as a React and TypeScript web application using server-side API routes for LLM calls. The frontend accepts pasted letter text and uploaded PDF or image files. Server-side endpoints handle letter analysis, translation, sentence explanation and product-support chat. Zod schemas validate API inputs and model outputs before content is displayed to the patient.

4.1. Letter analysis workflow

The analysis workflow takes either letter text or an uploaded file representation. The model is prompted to produce structured JSON containing: letter type, department or clinic, appointment date and time, location, contact information, clinician or team, action required, summary, checklist, questions to ask and missing-detail flags. A local fallback analyser generates a conservative response when the model is unavailable or unsafe.

The safety-bounded analysis workflow validates request length and input shape, generates a conservative local fallback, asks the LLM for admin-only structured JSON, validates the response against the schema, runs safety validation and applies rule-based unsafe-output filters. If the response contains unsafe clinical

advice or unsupported official claims, CareClarity returns the fallback response instead of displaying the model output.

4.2. Multilingual translation workflow

CareClarity includes translation support for selected community languages, including Bengali, Urdu, Arabic, Polish, Romanian, Punjabi, Hindi, Gujarati, Somali, Spanish, French, Chinese and Ukrainian. The translation endpoint is not designed to produce clinical advice. It translates administrative content and returns translation notes, important terms and a safety notice. The workflow checks for unsafe terms such as medication-change recommendations and falls back if such content appears.

4.3. Patient-facing outputs

The interface presents output as a dashboard rather than a free-form chat response. It includes a plain-English summary, extracted dates/times/locations/contact numbers, an action checklist, missing-detail warnings, appointment-readiness guidance, safe questions for the relevant NHS team and a family/carer summary.

5. Safety, Privacy and Trustworthy AI Controls

The central design choice is task narrowing. CareClarity does not attempt to answer medical questions. It instead directs users to GP practices, hospital teams, pharmacists, NHS 111 or emergency services where appropriate. This aligns with the principle that patient-facing AI systems should avoid creating a false impression of clinical authority.

The safety controls are layered: the system prompt restricts the task to administrative explanation, output must match a structured schema, safety validation classifies generated responses, rule-based filters detect medication-change advice or diagnosis claims, and unsafe or invalid outputs are replaced with conservative admin-only fallback guidance.

The prototype also avoids login and does not require a backend patient-record database. The LLM API key is kept server-side. These choices do not make the system clinically deployable, but they support a privacy-aware prototype architecture for evaluating patient-administration assistance.

6. Ethical Considerations and Reproducibility

CareClarity was developed and evaluated without using real patient letters, personal health records or identifiable NHS patient data. The benchmark documents are synthetic NHS-style administrative samples created to test system behaviour under controlled conditions. This design choice reduces privacy risk and allows the evaluation data to be inspected, reproduced and extended without exposing patient information.

The system is intentionally not a medical assistant, diagnostic tool or treatment-advice service. It does not recommend diagnoses, medicines, dosage changes, tests, procedures or clinical decisions. Its role is limited to administrative understanding: helping users identify dates, times, locations, contact routes, required actions and safe questions to ask the relevant NHS team. Any clinical uncertainty is directed back to qualified healthcare professionals, NHS 111 or emergency services where appropriate.

To support reproducibility and independent checking, the CareClarity project is publicly available at <https://github.com/imranalmunyeem/CareClarity> [12]. The repository can be used to inspect the prototype implementation, reproduce local tests and extend the benchmark with additional synthetic administrative letters.

7. Preliminary Offline Evaluation

This section reports a 100-document offline stress test of the implemented CareClarity prototype. The purpose of this run is to measure the deterministic local fallback analyser, schema validation and safety filters using controlled synthetic letters with deterministic ground-truth labels. No real patient data was used. The results should be interpreted as a system-verification baseline and error-analysis dataset, not as final evidence of deployed system performance.

7.1. Dataset and setup

The benchmark contains 100 labelled synthetic NHS-style administrative letters generated from controlled templates. Each document has known ground truth for date, time, location, contact number and expected

Table 2. 100-document offline benchmark dataset.

Category	Example content	Count
Appointment	Clear outpatient and video appointment letters	15
Referral / waiting list	Referral acknowledgement and waiting-list updates	7
Test instructions	Blood-test, scan and diagnostic appointment instructions	8
Prescription admin	Repeat-prescription collection and pharmacy admin notices	7
Clinic change	Relocation and hospital-initiated cancellation notices	14
Form / pre-assessment	Forms or questionnaire requests with deadlines	7
Missing-detail challenge	Missing date/time/location, incomplete phone number, conflicting instructions and action without deadline	42
Total	Synthetic NHS-style administrative documents	100

Table 3. Structured extraction performance for the 100-document offline benchmark.

System	Date/time	Location	Contact	Action checklist
CareClarity local fallback analyser	100.0%	79.0%	87.0%	100.0%

missing-detail flags. The templates cover routine appointment letters, diagnostic-test instructions, referral and waiting-list updates, prescription-administration notices, clinic changes, form/pre-assessment requests and explicit missing-detail challenge cases. Table 2 summarises the dataset composition.

7.2. Metrics

Field extraction was measured by comparing predicted values against manual labels. A field was marked correct when the expected date, time, location or contact number was present in the predicted structured output. If a field was intentionally absent from the document, the prediction was marked correct only if the system returned a missing-value response. Missing-detail detection was evaluated over six flag types: no date, no time, no location, unclear contact number, conflicting instructions and action required with no deadline.

For missing-detail detection, precision, recall and F1 are computed as:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (7.1)$$

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (7.2)$$

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (7.3)$$

Safety-boundary compliance was measured by scanning user-facing outputs for disallowed advice patterns, including medication-change advice, diagnosis claims and unsupported official NHS endorsement claims. Schema validity was measured using the implemented Zod response schema. Action output was counted as valid when the system generated at least one safe administrative checklist item and the safety status was **SAFE**.

7.3. Results

Table 3 shows structured extraction results for the local fallback analyser. Date and time extraction were correct for all 100 documents. Location extraction was 79/100 and contact extraction was 87/100. Action-checklist generation remained valid for all 100 outputs.

Table 4 reports missing-detail detection, safety and schema-validity results. The system produced 28 true positives, 73 false positives and 28 false negatives for missing-detail flags, giving precision 0.277, recall 0.500 and F1 0.357. All 100 outputs were schema-valid. No unsafe output was detected by the benchmark scanner.

Local implementation checks also passed: the full Vitest suite, benchmark run and TypeScript/Vite production build completed successfully, and all benchmark outputs passed Zod schema validation.

Table 4. Missing-detail detection, safety and schema-validity results.

System	Precision	Recall	F1	Unsafe	Schema-valid
CareClarity local fallback analyser	0.277	0.500	0.357	0/100	100/100

7.4. Error analysis and next evaluation

The stress test revealed a clear distinction between structured extraction and missing-detail flagging. Date/time extraction was robust on controlled templates because the fallback analyser recognises common UK patterns. Location and contact extraction were weaker for video consultations, cancellation wording, incomplete phone numbers and documents with no physical attendance site. Missing-detail precision was low because the heuristic over-flagged no-time and no-location in referral, waiting-list and form letters where no appointment had yet been booked; recall was limited by action-required-without-deadline and cancellation edge cases.

The next evaluation should compare a generic LLM prompt, CareClarity structured prompting without the safety validator and the full CareClarity workflow. Multilingual testing should separately check preservation of dates, times, locations, phone numbers, clinic names and action deadlines. This paper reports the offline fallback stress test only; broader LLM analysis and multilingual translation quality are not claimed as evaluated results here.

8. Discussion and Limitations

CareClarity demonstrates an approach to healthcare AI in which usefulness comes from narrowing the task rather than expanding model authority. Instead of attempting to replace a clinician, the system helps patients identify what a letter says, what practical information is present, what may be missing and which NHS team should be contacted. This makes the system easier to evaluate than open-ended clinical chat because many target fields are observable in the document.

The multilingual component is relevant in a UK healthcare context. It does not replace professional interpreting services, but it may help users obtain an initial administrative understanding and identify what to ask next. Evaluation should pay special attention to failure modes: a false missing-detail warning may inconvenience a patient, but false reassurance that a date, time or location is clear could be more harmful. A single unsafe medication recommendation should be treated as a serious failure.

Limitations remain. The benchmark is synthetic and may not capture the full variability of real NHS correspondence. The prototype has not yet been evaluated with patients, carers, NHS administrative staff or clinicians. Automated readability metrics do not guarantee comprehension, multilingual preservation checks do not guarantee culturally appropriate translation, and LLM behaviour can change across model versions. CareClarity should therefore be understood as a prototype research system, not a clinically deployed NHS service. Real-world deployment would require information governance review, usability testing, clinical safety review, security testing, accessibility testing and operational ownership.

9. Conclusion

CareClarity explores a practical middle ground for healthcare AI: patient-facing assistance that is useful because it is deliberately constrained. By focusing on administrative understanding rather than clinical advice, the system can extract key details, simplify wording, flag missing instructions and support multilingual access while maintaining explicit safety boundaries. The 100-document synthetic stress test provides a reproducible system-verification baseline and exposes missing-detail detection weaknesses for the next iteration.

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